

Adding Sound to your Models

By Tim Surtell



The 'playability' factor of many Meccano models can be considerably enhanced if they are capable of producing suitable sound effects. For example, a military model could emit the sound of gunfire, a model of an emergency vehicle could have a wailing siren, and a model steam locomotive could produce a 'chuff-chuff' sound.

Producing the sound of gunfire, or any other type of sound, would be tricky to do but for several special sound effects ICs which are available to easily do the job. Because these ICs have been designed for use in toys, they are easy to use and require minimal external components. Unfortunately, they do not all have the same facilities, and so they all have different application circuits.

Siren and Machine Gun Sounds

The UM3561 sound effect generator IC would be a good choice for models of emergency vehicles since it produces police car, fire engine, and ambulance siren sound effects. Which effect is played is determined by the connection of pin 6 to Vs, 0V, or nothing, as shown in the table below:

Sound Effect	Pin 6
Police Car Siren	No Connection
Fire Engine Siren	Vs
Ambulance Siren	0V

In addition, a pulsating machine gun sound effect, which could be useful in a military model, can be played at any time by connecting pin 1 to Vs.



Datasheets ► [UM3561 Sound Effect Generator IC](#) - PDF format (213kb)

The pin-out of the UM3561 is shown in figure 1. Pin 1 is the pin on the bottom left when the IC is held horizontal with the writing the correct way up. The little notch in one side also helps to identify pin 1, as does the small white dot next to this pin.

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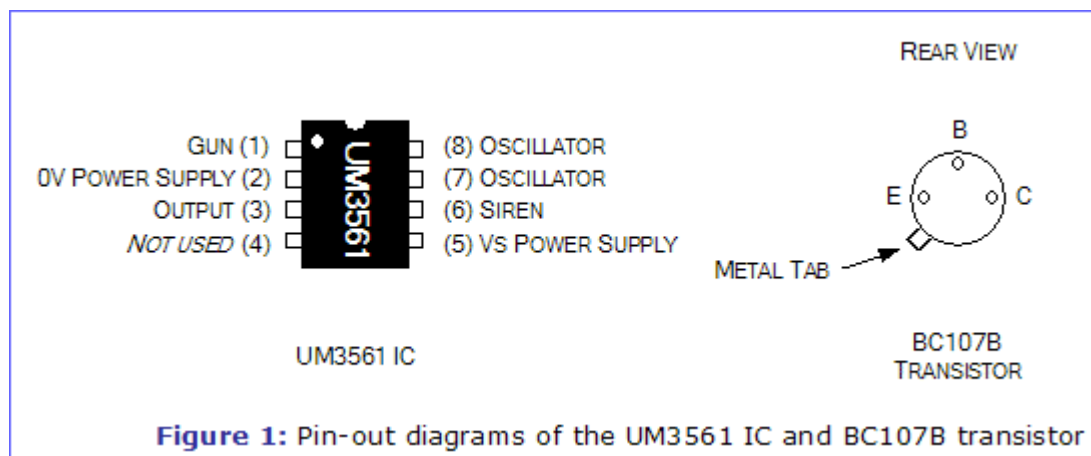
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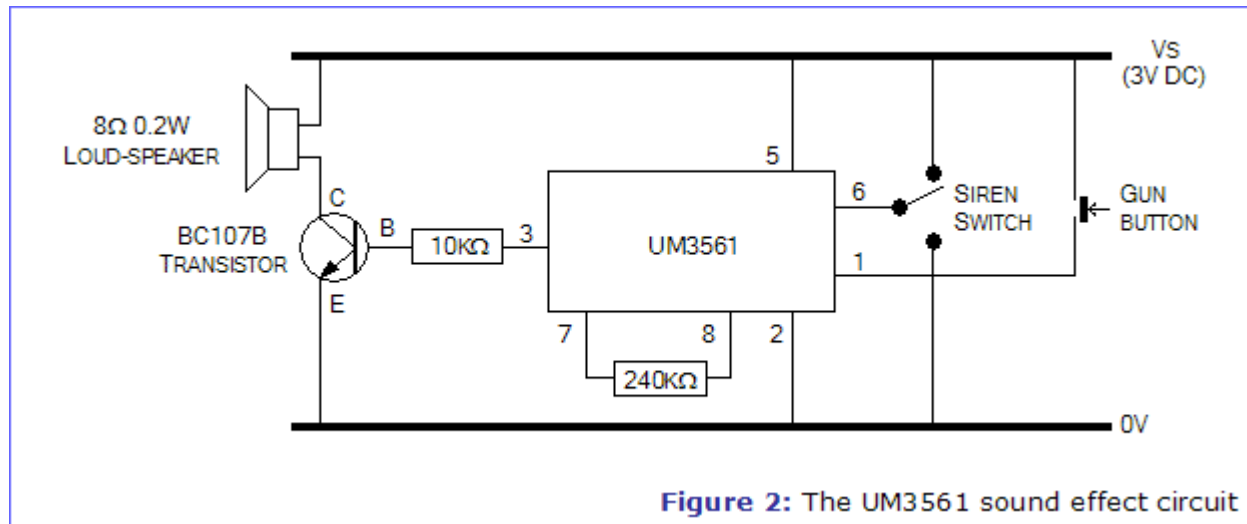
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The IC's power supply (3V DC) should be connected between pins 5 and 2. The sound effect output from pin 3 can be connected either to a piezoelectric sounder or to a small 8Ω , 0.2W loud-speaker via a BC107B transistor.

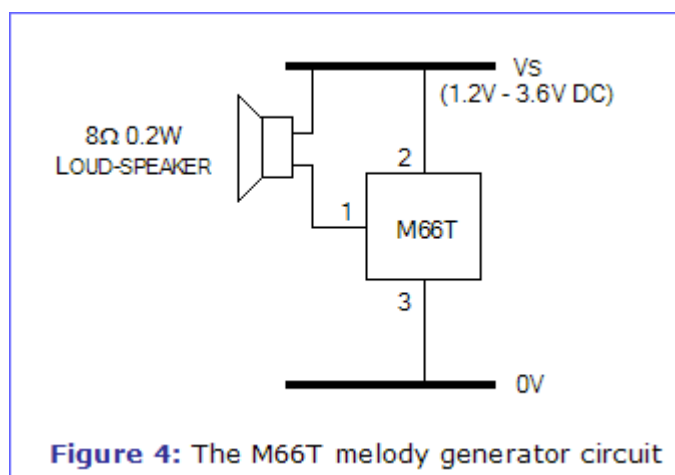
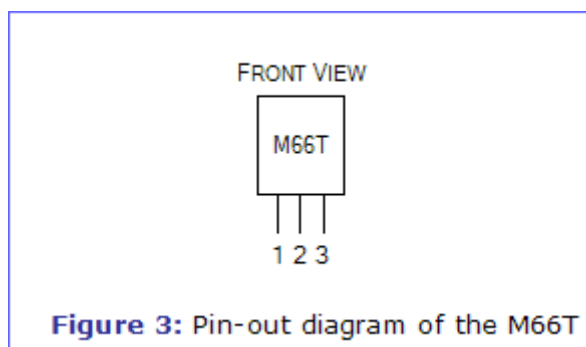
In the circuit diagram of figure 2, pin 6 has been connected to the common terminal of an SPDT centre-off switch so that any of the three siren sounds may be produced when the power is on. The machine gun sound effect can be played by pressing the Gun button when the power is on. Of course, should you only need to use one of the effects, you could hard-wire pins 1 and 6 as appropriate.



Playing a Tune

Several versions of the M66T melody generator IC are available, each of which plays a common tune such as "Happy Birthday" or "Jingle Bells". These ICs are powered from 1.2V to 3.6V DC, and are extremely easy to use: just connect up the power and a small 8Ω , 0.2W loud-speaker as shown in figure 4. No other components are required.

 **Datasheets** ► [M66T Melody Generator IC](#) - PDF format (50kb)

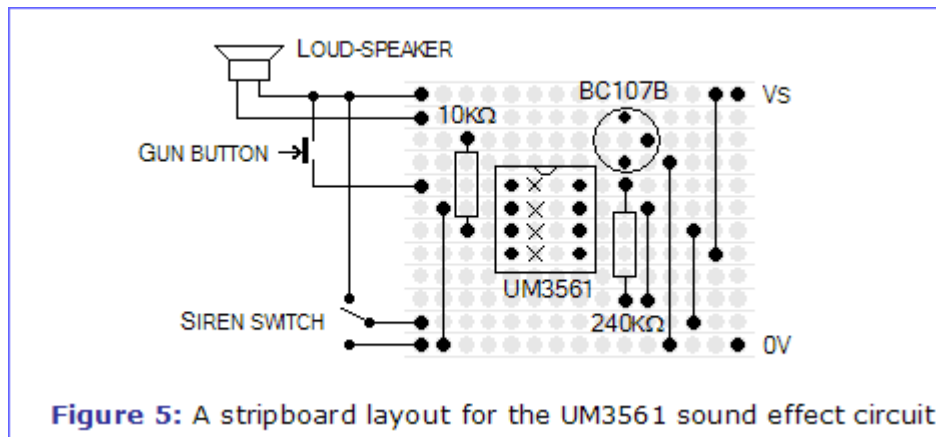


Building the Circuit

A stripboard layout for the UM3561 sound effect circuit shown in figure 2 is given in figure 5, and it will allow all four sound effects to be produced.

Construction

1. Cut a piece of stripboard to 15 tracks x 21 columns.
2. Fit the 5 wire links.
3. Fit the 10k Ω and 420k Ω resistors.
Fit the 8-pin IC socket, taking care to align the notch as shown in the layout. Do not insert the UM3561 at this time.
4. Fit the BC107B transistor. Place a crocodile clip around each leg of the transistor before soldering. This 'heat shunt' will prevent the heat from your soldering iron damaging the transistor. Try to keep the soldering time to a minimum for this component.
5. Insert the UM3561 into its socket, taking care to align the notch with the notch in the socket.
6. Cut the copper tracks where an X is shown.
7. Connect up the button, SPDT switch, and the loudspeaker as shown.
8. Connect up a 3V DC power supply and test the circuit.



Shopping List

Below is a list of the parts required to construct the UM3561 sound effect generator circuit on stripboard as shown in figure 5, except the UM3561 itself which cannot be purchased from Maplin, but is available from Rapid (see [links](#) page for contact details). Also given are the various versions of the M66T melody generator ICs.

Part Description	Quantity Required	Maplin Code	Unit Price	Total Price
10kΩ 0.6W Metal Film Resistor	1	M10K	£0.13	£0.13
240kΩ 0.6W Metal Film Resistor	1	M240K	£0.13	£0.13
45mm 8Ω 0.2W Loud-speaker	1	L68AA	£2.26	£2.26
8-pin Dual-in-line IC Socket	1	BL17T	£0.19	£0.19
BC107B Low Power NPN Transistor	1	QB31J	£0.81	£0.81
M66T Melody Generator IC ("Happy Birthday")	1	N16CG	£1.16	£1.16
M66T Melody Generator IC ("I'd Like To Teach The World To Sing")	1	GX58N	£1.16	£1.16
M66T Melody Generator IC ("It's A Small World")	1	N18CG	£1.16	£1.16
M66T Melody Generator IC ("Jingle Bells")	1	N23CG	£1.16	£1.16
M66T Melody Generator IC ("Looney	1	N12CG	£1.16	£1.16

Tunes" Theme)

M66T Melody Generator IC ("Old Macdonald Had A Farm")	1	GX57M	£1.16	£1.16
M66T Melody Generator IC ("Rock-a-Bye Baby")	1	N19CG	£1.16	£1.16
M66T Melody Generator IC ("Rudolph The Red Nosed Reindeer")	1	N22CG	£1.16	£1.16
M66T Melody Generator IC ("Star Trek" Theme)	1	N15CG	£1.16	£1.16
M66T Melody Generator IC ("Three Blind Mice")	1	N17CG	£1.16	£1.16
M66T Melody Generator IC ("Twinkle Twinkle Little Star")	1	GX55K	£1.16	£1.16
M66T Melody Generator IC ("Warning Tone")	1	GX60Q	£1.16	£1.16
M66T Melody Generator IC ("We Wish You A Merry Christmas")	1	N21CG	£1.16	£1.16
M66T Melody Generator IC ("White Christmas")	1	GX59P	£1.16	£1.16
M66T Melody Generator IC ("Winnie The Pooh" Theme)	1	N14CG	£1.16	£1.16
M66T Melody Generator IC ("You Are My Sunshine")	1	N20CG	£1.16	£1.16
Red Push-to-Make Button	1	FH59P	£1.16	£1.16
Stripboard SRBP (29 rows x 39 columns)	1	JP47B	£2.70	£2.70
Sub-miniature SPDT Toggle Switch Locking with Centre Off	1	FH01B	£2.20	£2.20

Grand Total £28.14

► [Full printer-friendly parts list](#)

Order the above items from Maplin via our Circuits Shop and help support the *Electronics in Meccano* website, without affecting the price you pay! Just click on an item for information about it and/or to add it to your Maplin order.

Orders can also be placed by telephoning Maplin on 0870 429 6000, or visiting their website at www.maplin.co.uk. Prices quoted are taken from the Autumn/Winter 2006/7 Maplin catalogue and include VAT at 17.5%.

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